

PRATIMAI · AI HANDOUTS

Decide, Evaluate, Design

My field book for judging AI systems, designing them responsibly — and defending the call

CLASS 10 · LEVEL 5 — EVALUATE & DESIGN

21 PAPER MISSIONS · 5 CHAPTERS · NO SCREEN NEEDED

PRATIMA AI Curriculum Team

How do I know? · Who is accountable? · What would change my mind?

THIS CASE FILE BELONGS TO

name · class · roll number

CASE OPENED ON ★ CASE CLOSED ON

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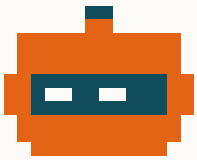
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PRATIMAI · AI HANDOUTS

Acknowledgments

To the Classes 6, 7, 8 and 9 students who carried this series on their shoulders — the detectives who counted pixel grids, the builders who audited guest lists, the analysts who voted with rulers and dice — this final book is yours. Your teachers reported the moments that turned practice into conviction: the day a class argued for a full period about which metric a screening tool should chase, the proposal that failed its own fairness check and survived anyway. Those reports became the chapters that follow.

To the curriculum reviewers who tested every mission against the promise that a board-year student needs **understanding, not another cram book**: thank you for holding the line. Nothing here asks you to memorise a syllabus; everything here makes the syllabus make sense. And to the adults who will be asked, over dinner, "who is responsible when a machine decides?" — this book trains a student to answer with a framework, an evidence table and an honest limit, and to keep their voice calm while doing it.

This series was built brick by brick by three teams working as one. The **AI research team** checked every explanation against how machines really learn, so nothing in this book is a myth wearing a friendly font. The **academic team** walked every chapter beside the school syllabus, so each game quietly trains a skill that examinations will ask for later. And the **child-psychology team** shaped the pace of every page: short missions for young attention, quick wins that open every chapter, confidence circles that make honesty feel safe, and space to write on every task — because an idea worked by the hand is remembered by the mind. Alongside them, curriculum reviewers guarded the "unplugged" promise: not one mission in this series asks for a phone, an account or a password.

Thanks go as well to the designers of Nunito and Baloo 2, whose friendly letterforms keep pages of hard argument feeling clear. Finally, to the AI Designer holding this copy: the last book of a series is a promise kept in public. Judge us gently, and send us what you find.

21 PAPER MISSIONS · 24 WORDS TO KEEP · 5 CHAPTERS · 1 PORTFOLIO EARNED

Before You Begin

Welcome back, Designer. Four years of evidence sit behind you: in Class 6 you **spotted** learning machines, in Class 7 you **operated** them, in Class 8 you **planned and defended** them, in Class 9 you explained **why they work and when they fail**. This year the questions grow up one last time: not just **how does it work** but **how good is it, who decides, and what should we build at all?** You will place every product you know on a model-family map, run a neural network with your own classmates as neurons, judge a medical screening tool with a confusion matrix, argue a triage case through four ethical principles, and design a complete, defensible AI system on paper — with a budget, an ethics review and a pitch. This is the year you take the AI Designer's seat for good.One honest sentence before you start. This is a board-exam year, and this book is **not** an exam-cram guide. It is something more useful: the reasoning underneath what exams examine. When you can judge a model with your own confusion matrix, the syllabus questions become arithmetic you have already lived. Keep your Class 9 book nearby — the maths there is the floor this book builds on.

WHY THIS BOOK WORKS THE WAY IT DOES

This handbook is built around how teenage brains actually learn. Every chapter opens with a quick win — motivation first, theory second. You **do** before you **define**: each idea is met through an experience before it is named, because memory hooks onto action. The confidence circles train metacognition — knowing how sure you are is a skill, and honest rating makes it stronger. Tasks are time-boxed because attention works in sprints; ideas return across chapters on purpose, because a second meeting is what turns exposure into understanding; and every task leaves space to write, because producing your own answer beats re-reading every time. Nothing here can be failed: a wrong answer you go on to check is a clue found early.

HOW TO READ A MISSION

T10-10 Mission title — PAIR · 15 MIN

- **QUICK WIN** sits at the right edge of the easier missions.
- **HANDS-ON LAB** marks the missions where you **build, measure, act or compute by hand** — the working labs of this book. There are eight. They are the fastest way to own an idea.

Every mission has a code like **T10-01** — **T10** means this Class 10 handout, and **01** is the mission number, so your teacher can say "open Task T10-11". This year the missions ask for **judgement with evidence**: an answer plus the reason, the number plus what it hides, the design plus its limits. In your mastery tracker, you rate yourself on four levels — **Emerging, Developing, Proficient, Advanced** — and you defend the level you claim. Nobody fails in this book; honest evidence always counts.

THE QUESTION HABIT — THE DESIGNER'S EDGE

Your five questions — **"How do I know?"**, **"What is missing?"**, **"Who made this?"**, **"Who is missing?"**, **"What is the trick here?"** — now take on the two that professionals are paid to ask. For every system you evaluate: **"Who is accountable when it fails — by name, in writing?"** And for every design of your own: **"What evidence would change my mind?"** A designer who cannot name what would change their mind is not designing; they are defending. Carry all seven questions into every debate, every review, and every rumour you meet this year.

DRILL THE SEVEN QUESTIONS — ON A SYSTEM IN THE NEWS

Pick one AI system from this week's headlines (a hiring filter, an exam proctor, a crop-advisory bot). Write one evidence-backed line per question — two of the seven are the accountability pair, and those two are the ones professionals are paid to ask:

1–3 **How do I know? What is missing? Who made this?** — one line on the system's evidence trail.

4–5 **Who is missing? What is the trick here?** — one line on whose voices the coverage leaves out.

6 **Who is accountable when it fails — by name, in writing?**

7 **What evidence would change my mind?** — the line that keeps you a designer, not a defender.

A Note for Adults

This handout builds **AI understanding** — how machines learn from data and how humans should judge them — through 21 paper-and-pencil missions. It is fully **unplugged**: no task requires a device, an account, a photo, a voice recording or any personal information. This year the big ideas are model families (rule-based, supervised, unsupervised, reinforcement — and neural networks as layers of weighted votes); how machines see (pixels, convolution) and read (text normalisation, bag-of-words, TF-IDF); honest evaluation (train/test splits, confusion matrices, precision, recall and choosing a metric for the stakes); ethics as practice (the four bioethics principles, algorithm audits, privacy rights, generative-AI integrity); and a complete capstone design brief with an ethics gate. Where the book touches India's data-protection law or board syllabi, a **verify-current-rules** note sits in the teacher pack — regulations move faster than print. The best help you can give a student in an exam year is to treat this reasoning as the subject itself: ask for the evidence, the metric choice, and the limit in the same breath.

Four ways to help — without giving answers

1 Ask for the metric

When your student says “the model is 99% accurate”, ask **“accurate at what — and who pays for the errors?”** Metric choice is the year’s core skill; practise it at the dinner table.

2 Respect the argument

Swap “good answer” for **“you changed your mind in front of evidence”** — the rarest skill this book teaches is updating under pressure, and it deserves to be named.

3 Protect the portfolio

The capstone design brief is a portfolio piece. Keep it flat, dated and un-marked: what a student can **show and defend** outranks what they once scored.

4 Connect to real governance

Privacy rules, AI guidelines and board curricula are all in motion. Read one real notice or policy together — the book trains the reading, the news provides the text.

SIGNS OF DESIGNER GROWTH WORTH PRAISING

Watch for these quiet changes — they matter more than any correct answer: your student **names the metric** before naming the score · they say **“it depends on the cost of each error”** and mean it · they **change their position** when evidence lands, and say why out loud · they ask **“who is accountable, by name?”** of systems in the news · their design brief contains a limit they discovered themselves. When you spot one, name it: “That is exactly what designers do.”

IF YOUR CHILD ASKS…

“Is this book enough for the AI syllabus?” — Say: it deliberately teaches the reasoning the syllabus examines — model families, evaluation, ethics frameworks, the design process — without duplicating any board textbook. For the printed syllabus detail, use the official document with a teacher. **“Will AI take the careers I’m aiming for?”** — Say: this book’s honest answer is Chapter 5's — some tasks automate, most roles reshape, and the durable skills are exactly what the capstone practices: scoping, judging evidence, and taking responsibility. **“Why so much paper in a programming year?”** — Say: because the professionals design on whiteboards first; and because on paper, nothing hides inside a library call — every weight, every metric, every gate is visible and yours.

My AI Journey Map

THE FIVE STRANDS YOU WILL MEET AT EVERY STOP

U Understand AI
Which model family is this — and what exactly does 'deep' add?

D Data
Whose data, whose consent — and what does privacy law owe me?

L Learn
How do I judge a model honestly — and which metric do the stakes demand?

W World
How do machines see and read — and what does generative AI owe me?

R Responsibility
Who is accountable, and how do I design so the answer exists?

THINKING-TOOLS THREAD

runs through every stop: **decompose** the judgement · **compute before you claim** · **argue the counter-case** · **design the gate before the deployment**

THE FIVE STOPS ON YOUR FINAL CASE

1. **Models Explained** — *families · learning types · weighted votes*
2. **Machines That See and Read** — *pixels · convolution · TF-IDF · bots*
3. **Judging a Model** — *splits · confusion · precision · base rates*
4. **Ethics, Governance & Society** — *principles · audits · privacy · careers*
5. **Responsible AI Design Brief** — *the full pipeline · pitch · your map ahead*

BY THE END OF THIS BOOK, I CAN…

- [ ] place any AI product on the model-family map — rule-based, supervised, unsupervised or reinforcement
- [ ] explain a neural network as layers of weighted votes — and run one by hand
- [ ] split data into train and test sets honestly, and explain why leakage ruins a score
- [ ] build a confusion matrix from raw results and compute accuracy, precision and recall from it
- [ ] choose the right metric for a context — and defend the choice with the cost of each error
- [ ] explain the base-rate trap and why a “99% accurate” claim can still be almost useless
- [ ] apply a 3×3 kernel to pixels by hand and sketch what convolution layers do
- [ ] normalise text, build a bag-of-words table, and tell script bots from smart bots
- [ ] run the four ethical principles over a real AI case — and find where each one bites
- [ ] design a complete AI project on paper: scope, data, model, evaluation, deployment, ethics gate, pitch

MY BOARD-YEAR RHYTHM — THREE PERIODS, THREE PROMISES

This is a board year, and the designer’s edge is rhythm, not cramming. Write the promise you are making for each stretch of the year — reread these lines at every chapter checkpoint:

APR–JUL foundations I will not fake: _____

AUG–NOV practice I will do under time, honestly: _____

DEC–MAR revision I will run from my own notes, not borrowed ones: _____

DAY 1 My starting line — a pledge to the standard

2 MIN

Fill this in before Chapter 1. The Designer's Certificate at the back will ask for the evidence beside these lines.

One AI system in my life I intend to be able to **judge** — not just use — by June:

The metric-vs-cost question I expect to argue about this year, and with whom:

My integrity clause — the line I will not cross with generative tools, in my own words:

Signed: _____ Date: _____

HOW TO USE THE RATING SCALE — THE DESIGNER'S READING HABIT

Rate with **evidence**, not mood: a level you cannot demonstrate in the back tracker is a level you do not own yet. Designers rate low on purpose and show the fix — that habit, more than any single answer, is what the board year rewards. Re-rate after every chapter checkpoint; keep the first circle visible so the distance stays honest.

CARD The designer's seven questions — my pocket copy

1–5 **How do I know? What is missing? Who made this? Who is missing? What is the trick here?**
6 **Who is accountable when it fails — by name, in writing?**
7 **What evidence would change my mind?** — the line between a designer and a defender.

The system I will run all seven on first: _____

BEFORE EVERY CHECKPOINT — MY THREE RE-RATING HABITS

1. Re-rate with evidence, not mood — show the mission that proves the level. **2.** Keep the old circle visible; the distance is the point. **3.** One line on what changed my rating — the failure, the fix or the question.

MY MODEL SHELF — THREE PRODUCTS I WILL JUDGE FIRST

For Chapter 1: name three AI products you actually use, and one data question you will ask of each when the family map opens:

1. _____
data Q _____
2. _____
data Q _____
3. _____
data Q _____

YOUR HONEST-RATING SCALE — THIS YEAR AND BEYOND

Level	It means — exactly	The evidence that earns it
Emerging	I have met it and can follow someone else using it.	a completed mission with help
Developing	I can do it with my notes open.	a re-run mission, notes beside me
Proficient	I can do it cold — and explain it to someone else.	a mission done clean, then taught home
Advanced	I can teach it, break it on purpose, and defend its limits.	a checkpoint answer with the counter-case

Rate yourself with evidence in the back tracker — a level you cannot show is a level you do not own yet. Designers rate low and show the fix; that is the whole habit.

MY STARTING LEVELS — CIRCLE TODAY'S HONEST GUESS, SHADE THE END-OF-YEAR TRUTH

THE SKILL	TODAY	YEAR-END
place a product on the model-family map	E · D · P · A	E · D · P · A
run a neural network by hand	E · D · P · A	E · D · P · A
judge a model with a confusion matrix	E · D · P · A	E · D · P · A
apply a kernel to pixels by hand	E · D · P · A	E · D · P · A
build a bag-of-words table	E · D · P · A	E · D · P · A
run the four principles on a real case	E · D · P · A	E · D · P · A
design and pitch a full brief, gates shut	E · D · P · A	E · D · P · A

Be generous with TODAY and ruthless at YEAR-END — the gap between the two columns is the course.

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The boxes in this book

You will meet these boxes on every case. Each one wants something different from you — collect them all.

MISSION

A paper mission with a code like **T6-01**. The label tells you who you work with and how many minutes you get. Do it — then write. Writing **is** the mission.

WORD POWER

A word worth keeping in your designer kit. Read it three times today — words are the tools of thinking.

SELF-CHECK

Skills to grade yourself on. Shade a confidence circle honestly — honest circles make you learn faster.

THINK & INK

A question with lines. There is no single right answer here — your **reasons** are the answer.

Also inside: **MYTH BUSTER** busts a wrong idea many people believe · **DESIGNER’S NOTE** hands you a professional’s secret · **HOME LINK** (dashed) is a mission to take home and teach an adult.

1. Models Explained

GUIDING QUESTION

What exactly is the machine doing under there?

Every AI system you have met since Class 6 — spam filters, recommenders, voice assistants, screening tools — lives somewhere on one map. This chapter draws the map. You will sort ten real products into model families, meet the three great learning strategies (supervised, unsupervised, reinforcement), run all four learning jobs (classification, regression, clustering, association) on a kirana-store case, and then build a neural network the honest way: with your classmates as neurons, weights on paper, and a vote you can watch. No black boxes survive this chapter — and the ones that pretend to, you will now be able to name.

MISSIONS T10-01 – T10-04
OUTCOMES 10.U1 · 10.L1 **STRANDS** U L

Links: Maths — functions & relations · Science — neuron models · Economics — market behaviour

In this chapter

1.1 The model family map 12
1.2 Three ways to learn — same examples, three different conversations 14
1.3 The four learning jobs 15
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TERMS YOU'LL DESIGN WITH

model family · supervised learning · unsupervised learning · reinforcement learning · weight

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

Your phone suggests "University" after "Delhi". One honest line: did it LEARN that from examples, or was it typed by hand? Keep the answer — page 4 settles it.

DESIGNER'S TOOLKIT — YOU WILL NEED: pencil · ruler · 10 slips of paper

1.1 The model family map

Last year's great divide — rule-based versus learning-based — was the first cut. This year we make the map precise enough to place anything on it. **Rule-based** systems follow logic a person typed: every behaviour is visible, testable, and frozen until a human rewrites it. **Learning-based** systems tune themselves from examples — and the learning trunk splits three ways. In **supervised** learning, every training example arrives with the correct answer attached, so the system learns the link from input to known output. In **unsupervised** learning, no answers are supplied; the system searches for structure — groups, co-occurring pairs — that nobody labelled in advance. In **reinforcement** learning, the system acts, receives a reward or penalty, and adjusts its strategy over many rounds, like learning a game by playing it. Deep learning, the word behind most headlines, is not a fourth family: it is a **technique** for building the learning trunk with many stacked layers — which is why it inherits both the power and the data-hunger of that trunk.

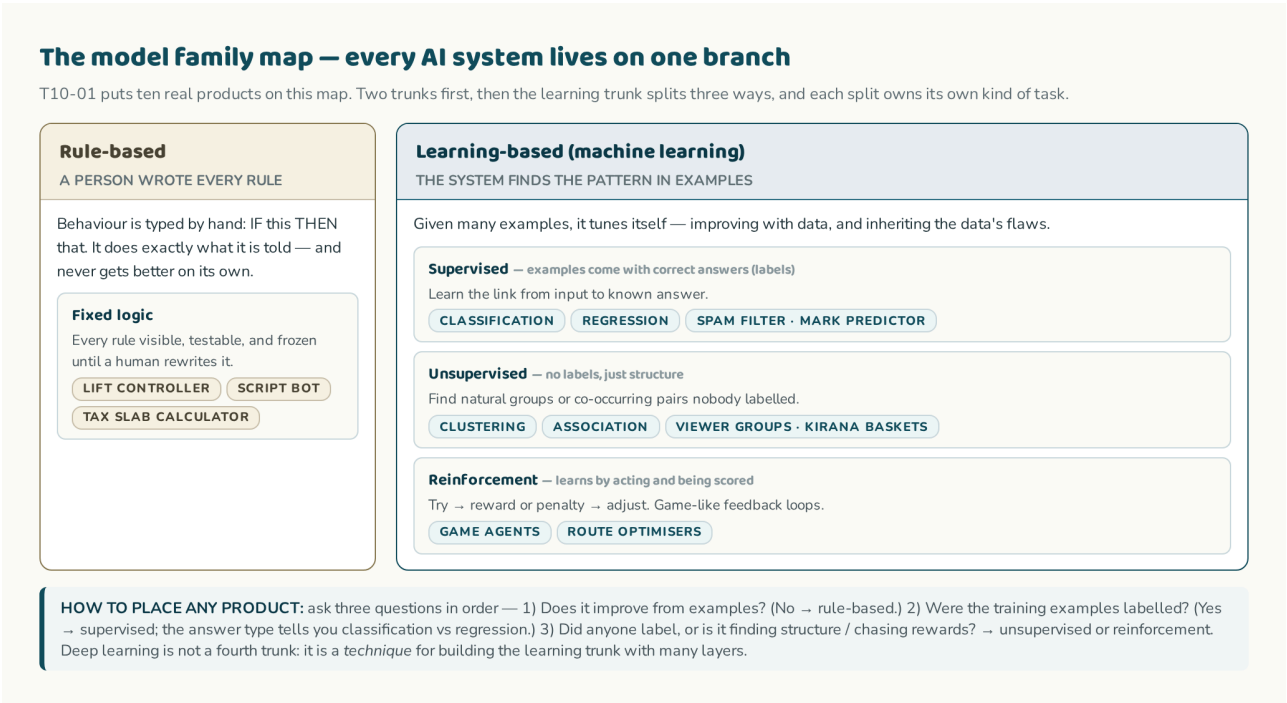


FIG The model family map — two trunks, three branches, four jobs. T10-01 puts ten real products on it.

T10-01 Model Map

■ QUICK WIN PAIR · 15 MIN

Ten products sit in the list below. For each one: **which family, which branch — and what would have to be true about its training data?**

1. a lift with floor buttons

family / branch / one data question

2. a spam filter

family / branch / one data question

3. an essay-writing chatbot

family / branch / one data question

4. a video app's 'people like you' feed

family / branch / one data question

5. a chess program that improves by playing itself

family / branch / one data question

6. a weather forecaster that estimates tomorrow's rainfall in mm

family / branch / one data question

7. a customer-support bot that only follows scripted menus

family / branch / one data question

8. a tool that groups library books by topic, with no labels given

family / branch / one data question

9. a kirana store's 'customers who bought rice also bought dal' rule

family / branch / one data question

10. a game agent that learns from points and penalties

family / branch / one data question

The two hardest calls: which pair of products will your class argue about most — and what single extra piece of information would settle it?

WORD POWER · 1

Definition: **rule-based system**

A system whose every behaviour follows rules written entirely by people.

Read it three times today — it sticks!

WORD POWER · 2

Definition: **supervised learning**

Learning from examples that come with correct answers attached.

Read it three times today — it sticks!

1.2 Three ways to learn — same examples, three different conversations

The three branches are not three technologies; they are three **conversations with the data**. Supervised learning asks: “here is the question and the answer — find the connection.” Unsupervised learning asks: “here is a pile of raw examples — tell me what structure hides inside.” Reinforcement learning asks: “here is a world that scores your moves — find a strategy.” The same classroom data can feed all three, and each returns a different kind of useful. This is why professionals name the branch **before** collecting data: the branch decides what data is worth having, what “correct” means, and what the system can never tell you.

T10-03 Scenario Sort

GROUP · 15 MIN

Six scenarios are on the cards your teacher holds (or below). For each: **supervised, unsupervised, or reinforcement — and the one sentence that justifies it.**

Scenario	My call	The deciding sentence
A tuition app marks practice answers right or wrong, then predicts which topics a student will fail		
A city bus planner finds that weekend trips naturally fall into five rider groups it never named		
A delivery drone learns routes by earning points for speed and losing them for crashes		
A clinic labels 5,000 X-rays ‘healthy’ or ‘needs review’ and trains a checker		
A music app discovers overnight that listeners of one singer often also save a second singer		
A robot vacuum improves its room-coverage path as battery penalties and praise accumulate		

The trap to discuss: two of these scenarios could plausibly run on a different branch. Which two — and what would change in the data collection if we switched?

WORD POWER · 3

Definition: unsupervised learning

Finding structure in examples that carry no labels at all.

Read it three times today — it sticks!

WORD POWER · 4

Definition: reinforcement learning

Learning a strategy by acting, then adjusting to rewards and penalties.

Read it three times today — it sticks!

1.3 The four learning jobs

Inside the branches, systems do one of four **jobs**. **Classification** outputs a category: spam or not, healthy or needs-review. **Regression** outputs a number on a scale: tomorrow's demand in crates, marks to expect. **Clustering** outputs groups nobody pre-named: your class sorted by sleep habits, riders sorted by trip shape. **Association** outputs pairings: people who buy rice also buy dal. Classification and regression are supervised jobs; clustering and association are unsupervised jobs. When you meet a product, naming its job is the fastest route to its limits — a classifier has never heard of a number, and a regression will happily estimate a category if you let it.

T10-04 Market-Basket (Kirana Store) Associations

PAIR · 15 MIN

The Karim kirana store's last 8 bills:

Bill	Items bought together
1	rice, dal, oil
2	rice, dal, soap
3	rice, dal
4	bread, jam
5	rice, soap, oil
6	bread, jam, milk
7	rice, dal, soap
8	bread, milk

a) Count pair frequencies. Which two-item pair is the strongest association, and what is its confidence as a percentage of bills containing the first item?

b) A sales app suggests “add soap” to every rice purchase. How often would it be wrong, using these bills?

c) What is the **smallest change** to bill 3 or 4 that would break the rule — and does that worry you about rules built on 8 bills?

d) Name one shop decision this association supports — and one it must NOT decide alone.

WORD POWER · 5

Definition: **classification**

A learning job whose output is a category from a fixed list.

Read it three times today — it sticks!

WORD POWER · 6

Definition: **regression**

A learning job whose output is a number on a scale.

Read it three times today — it sticks!

WORD POWER · 7

Definition: **clustering**

A learning job that finds groups nobody named in advance.

Read it three times today — it sticks!

WORD POWER · 8

Definition: association

A learning job that finds items that keep appearing together.

Read it three times today — it sticks!

1.4 Neural networks: layers of weighted votes

Strip away the mystique and a neural network is arithmetic you can run. Each **neuron** multiplies its inputs by numbers called **weights**, adds the results, and passes the total through a simple squashing rule. Neurons sit in **layers**; each layer's outputs become the next layer's inputs. Training is nothing more mysterious than the slow search for better numbers in the lines. The power comes from stacking: early layers answer tiny questions (an edge in one corner of a photo, a frequent pair of syllables), and deeper layers combine those answers into bigger ones (a face, a phrase, an intent). The costs are just as real: deep stacks need enormous data, their reasoning resists simple explanation, and their confidence is not conscience.

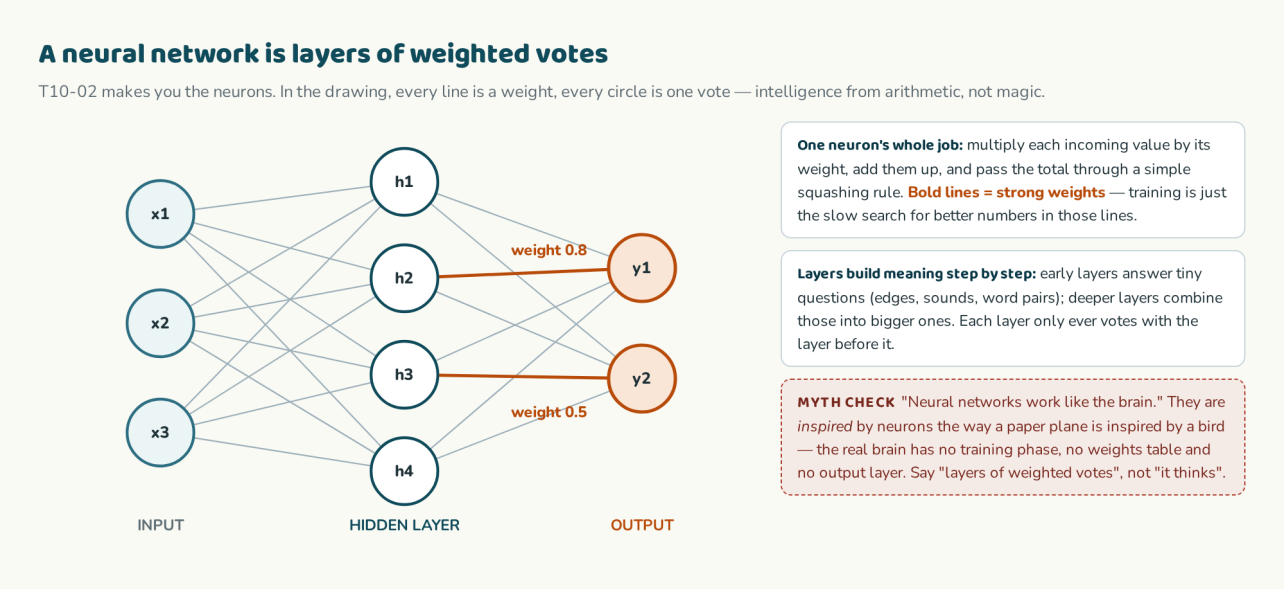


FIG One small network. Every line is a weight; bold lines dominate the vote. Training = searching for better numbers in the lines.

T10-02 Human Neural Network

HANDS-ON LAB CLASS · 30 MIN

Your class becomes the network. Three students are **inputs** holding a number card each (0 to 5). Four students are the **hidden layer**. Two are **outputs**.

Round 1 — the forward pass. Each hidden student picks one weight (0 to 1) per input, multiplies, adds, and writes the total. Output students repeat the step. **Record the whole pass in the table:**

connection	h1	h2	h3	h4	out
x1 · value, weight, product	x2 · value, weight, product	x3 · value, weight, product	hidden total passed on		

Round 2 — train it. The class sets a goal ("output y1 should exceed y2 for today's inputs"). Adjust ONE weight per trial. Run the pass again. How many trials to reach the goal?

Round 3 — the honest verdict. What did the network do that a single if-then rule could not — and what did it do **worse** than the rule?

WORD POWER · 9

Definition: **weight**

The adjustable number a neuron multiplies an input by — the 'strength' of a connection.

Read it three times today — it sticks!

WORD POWER · 10

Definition: **layer**

A row of neurons whose outputs feed the next row — depth comes from stacking.

Read it three times today — it sticks!

■ MYTH BUSTER

MYTH *"More layers always means smarter."*

TRUTH More layers mean **more combinations**, not more sense. Depth helps only when the data is rich enough to feed every layer — otherwise deeper networks memorise noise and confidently mislabel. The honest sentence: layers buy **capacity**, and capacity must be paid for in data.

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can place any product on the model-family map and defend the placement with one data question ○○○
- [ ] I can tell supervised, unsupervised and reinforcement apart by what each asks of its data ○○○
- [ ] I can name all four learning jobs and match each to a decision a shop or school could make ○○○
- [ ] I can run a forward pass by hand and explain what a weight does to the vote ○○○
- [ ] I can say what depth buys a network — and what it costs ○○○

THINK & INK

A system in my life I once called "smart". After this chapter, its honest description is a family, a branch and a job — they are …

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I sorted machines by how impressive they sounded. Now I sort them by …

CHAPTER 1 CLUES — POCKET THESE

- **Two trunks first:** rule-based (a person wrote it) versus learning-based (it found the pattern). Every other word on the map is inside the learning trunk.
- **Three conversations with data:** supervised (answers attached), unsupervised (no labels), reinforcement (scored moves). Name the branch before collecting.
- **Four jobs:** classification (a category), regression (a number), clustering (un-named groups), association (appearing-together pairs).
- **A neural network** is layers of weighted votes — power from stacking, cost in data and explainability.

The clue I would tell my family tonight:

Chapter 1 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. A chess app “that got better the more it played” and a lift controller are both ‘AI’ on your map. Name the placement of each — and the one question that separates them.

2. The clinic X-ray checker (T10-03) is supervised. What exactly were the labels, and what could the system never learn from that dataset?

3. In your human neural network, output y1 beat y2 after one weight changed. Explain — in the language of votes — why that small change could flip a decision.

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT MODEL FAMILIES

MODEL MAP RE-CHECK — TWO FRESH PRODUCTS, NO PEEKING

A queue-scanning app at the temple prasad counter

family · branch · one data question:

A cricket coach app that learns your bowling action

family · branch · one data question:

2. Machines That See and Read

GUIDING QUESTION

What does a machine actually receive — and what does it do with it?

Class 7 taught you that a camera sees a grid of numbers. This year you learn what machines **do** to those grids: you will run a real 3×3 kernel across a six-by-six image by hand and watch an edge announce itself, sketch how stacked filters become feature maps, then turn to reading — cleaning text until a machine can count it, building bag-of-words and TF-IDF tables with your own pencil, and drawing the line between a script bot that recites and a smart bot that predicts. Seeing and reading turn out to be the same trick: turn the messy world into numbers, then weight the numbers.

MISSIONS T10-05 – T10-09
OUTCOMES 10.V1 · 10.N1 **STRANDS** W

Links: Maths — matrices & weighted sums · Science — optics & the eye · Languages — morphology, root words

In this chapter

2.1 Pixels: the only thing a camera ever gives you 22
2.2 From edges to objects: the idea of a CNN 23
2.3 Teaching machines to read: clean the text first 25
2.4 Bag-of-words and TF-IDF: counting your way to meaning 26
2.5 Script bots versus smart bots 27

TERMS YOU'LL DESIGN WITH

pixel **grayscale** **kernel** **feature map** **normalisation**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

Draw the letter 'A' twice — once neat, once rushed. Write one line: what stayed the same in the **numbers** if a machine measured dark and light? Keep your answer for Section 2.2.

DESIGNER'S TOOLKIT — YOU WILL NEED: pencil · ruler · calculator (optional — all sums are small) · one printed or drawn 6×6 grid per pair

2.1 Pixels: the only thing a camera ever gives you

A greyscale image is a rectangle of numbers: 0 for pure black, 255 for pure white, and the greys between. Colour triples the story — every position stores three numbers for red, green and blue, and mixing those three channels in different amounts is how every screen you have ever watched makes every colour it has ever shown. Nothing enters the model except these numbers: no “cat-ness”, no context, no memory of the real scene. This is why machines can be fooled by things eyes forgive — a sticker on a sign, a tinted screen, a photo taken at an angle. The numbers changed; to the machine, the world **did** change.

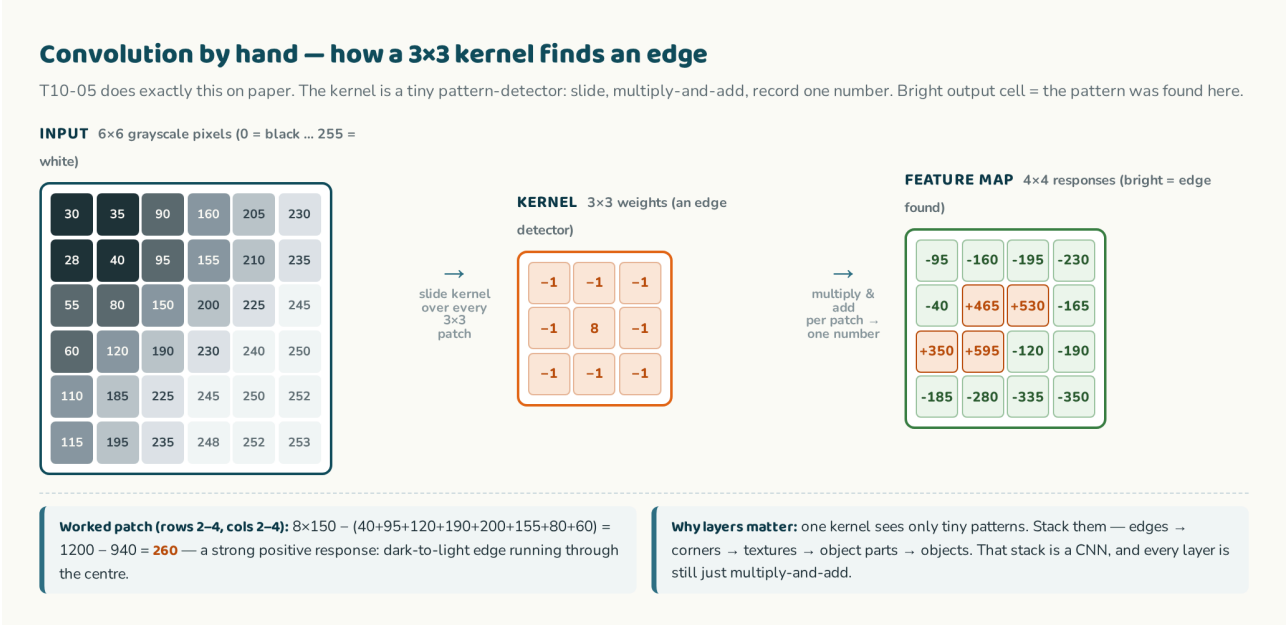


FIG The full convolution move you are about to do by hand: kernel in, multiply-and-add, one response number out. Bright cells mean “pattern found here”.

T10-05 Pixel Edge Hunt

■ HANDS-ON LAB PAIR · 25 MIN

The input grid below is a 6×6 greyscale patch with a dark-to-light edge running down the middle. Your kernel is the 3×3 edge detector from the figure: −1 on the border, 8 at the centre.

30	35	90	160	205	230
28	40	95	155	210	235
55	80	150	200	225	245
60	120	190	230	240	250

KERNEL

−1	−1	−1
−1	8	−1
−1	−1	−1

MY 4×4 ANSWERS

Procedure — twice by hand, then you may sprint:

1. Place the kernel over the top-left 3×3 patch. Multiply each pixel by its weight and add all nine products. Write the response in your 4×4 grid.

2. Slide one cell right. Repeat. One full row takes five moves.

110	185	225	245	250	252
115	195	235	248	252	253

3. Finish all 16 responses. **Where do the big positive numbers appear?** Mark that column.

4. Swap grids with another pair and check one response each — argue any difference to the arithmetic.

What the hunt proves: the kernel never saw the "edge". It only multiplied and added — and the edge announced itself anyway. That is a feature detector: no understanding, just arithmetic that fires when a pattern lines up.

Prediction before you run it: where will the strongest response appear — and why there, not one column to the left?

WORD POWER · 11

Definition: **pixel**

One position in an image grid, holding a number for brightness — three numbers for colour.

Read it three times today — it sticks!

WORD POWER · 12

Definition: **grayscale**

An image where each pixel holds a single brightness number from 0 to 255.

Read it three times today — it sticks!

2.2 From edges to objects: the idea of a CNN

One kernel finds one kind of pattern. A convolutional network stacks the idea: dozens of kernels find edges and blobs, the next layer combines edges into corners and textures, the next combines those into eye-shapes and wheel-shapes, and deeper layers answer the questions we care about — "is this a plant leaf or a diseased leaf?" Each layer's outputs are called a **feature**

map: a new grid of numbers recording how strongly each pattern fired everywhere in the image. Nobody writes the kernels by hand any more; training discovers them. But the arithmetic you did in T10-05 is exactly what every one of those kernels does, billions of times.

T10-06 Feature-Map Sketch

ALONE · 15 MIN

A 5×5 image contains: a vertical dark edge on the left, a bright square top-right, and smooth grey elsewhere.

a) Sketch the feature map for a **vertical-edge kernel** — which cells fire strongly, which stay near zero? Mark them.

b) A second kernel hunts **bright corners**. Sketch its feature map — different cells now.

c) The next layer receives BOTH maps. What can it answer that either map alone could not?

d) Your T10-05 kernel knew nothing about leaves. What, then, does a "plant disease classifier" actually have to be given — and by whom?

WORD POWER · 13

Definition: kernel

The small grid of weights slid across an image — one learned pattern-detector.

Read it three times today — it sticks!

WORD POWER · 14

Definition: feature map

The grid of responses one kernel produces — where its pattern fired.

Read it three times today — it sticks!

2.3 Teaching machines to read: clean the text first

A machine cannot count “The”, “the” and “THE” as the same word until someone makes them the same. **Normalisation** is the laundry list: lowercase everything, strip punctuation, split off stop-words (the, is, at — words so common they carry no topic), and chop words to their stems (study, studies, studying → **studi-**). Boring, yes — and decisive: skip it, and half your counts are duplicated noise. Every serious text system, from spam filters to translation engines, begins with this same scrub.

T10-07 Text Cleaner

HANDS-ON LAB · PAIR · 15 MIN

Normalise these four messages exactly as a machine would — lowercase, punctuation stripped, stop-words marked out, stems chopped:

	Message
M1	URGENT!!! Your prize is WAITING — claim your FREE prize now
M2	The class test is urgent; the prize-winning project is due tomorrow
M3	FREE workshop: studying smarter, not longer — free entry
M4	Mom says dinner is waiting, don’t study late tonight

Write each cleaned version here (stems with a dash):

M1 CLEAN

M2 CLEAN

M3 CLEAN

M4 CLEAN

The judgement call: “urgent” appears in M1 and M2, but means something different in each. What information did normalisation **throw away** — and how might that hurt a spam classifier?

WORD POWER · 15

Definition: **normalisation**

Cleaning text so machines can count it: lowercase, no punctuation, stop-words out, stems chopped.

Read it three times today — it sticks!

2.4 Bag-of-words and TF-IDF: counting your way to meaning

After cleaning, a message becomes a **bag of words** — order lost, counts kept — placed side by side with every other message in a table: rows are messages, columns are words, cells are counts. From that table, **TF-IDF** asks the sharpening question: which words make THIS message different from the others? Term frequency says how often a word appears here; inverse document frequency discounts words that appear nearly everywhere (like "free" in a folder where every third message shouts it — or is it the reverse?). The arithmetic is one division and one logarithm you need not compute precisely — what matters is the **ratio logic**: common-in-this-message, rare-everywhere-else, means informative; common everywhere means nearly worthless.

T10-08 Bag-of-Words & TF-IDF by Hand

HANDS-ON LAB PAIR · 25 MIN

Use the four **cleaned** messages from T10-07 (or the provided versions if your teacher hands them out).

a) Build the bag-of-words table: one row per message, one column per distinct content word, counts in the cells. (Choose your 8–10 columns as a class so all tables match.)

BAG-OF-WORDS TABLE — ROWS M1–M4, YOUR CHOSEN COLUMNS

b) Circle the word with the **highest document frequency** — in how many messages does it appear? What is its value for telling messages apart?

c) Underline the word with the **highest TF-IDF intuition** — frequent in one message, rare elsewhere. Which message does it identify?

d) M2 contains “urgent” sincerely; M1 uses “URGENT!!!” as bait. Both bags contain **urgent** once. What did the bag-of-words model lose — and name one feature a smarter model might add back.

e) A spam filter trained on these four messages gets a new one: “FREE FREE FREE prize claim now!!” Predict its label from your table, and give the table cell(s) that decide it.

WORD POWER · 16

Definition: **bag-of-words**

A table of word counts per message — order lost, frequency kept.

Read it three times today — it sticks!

WORD POWER · 17

Definition: **TF-IDF**

A score that lifts words frequent here but rare elsewhere — the informative ones.

Read it three times today — it sticks!

2.5 Script bots versus smart bots

Not every chatbot is an AI system, and the difference is testable. A **script bot** walks a tree a human drew: “press 1 for balance, press 2 for statements” — or its chat-window cousin, matching keywords to canned replies. Ask it something off the tree and it fails the same way

every time. A **smart bot** predicts: it scores likely continuations from patterns in text it learned from, which is why it handles novelty — and why it can answer fluently and wrongly at once. The exam-style question is now an engineering one: **which does this task deserve?** Script bots are auditable, predictable and safe where the domain is closed; smart bots flex where language wanders — at the price of inventing, and of needing everything you will build in Chapter 3 to be judged honestly.

T10-09 Script Bot vs Smart Bot

GROUP · 20 MIN

Draw both flow diagrams for a **school office enquiry bot** (admission dates, bus routes, fee queries, lost property).

DRAWING A — SCRIPT BOT

Boxes and arrows only: greeting → question type → the exact branches you would write → answers → handoff. Every branch drawn; nothing undrawn can happen.

DRAWING B — SMART BOT

One box that predicts, then the safety rails around it: what it must refuse to answer, when it must hand to a human, how its answers get checked.

The design decision: for THIS office, which bot would you ship — and name the one question a parent could ask that breaks your choice?

Group verdict: one argument for script that your group could not refute, and one for smart. Keep both — they return in the Chapter 5 brief.

MYTH BUSTER

MYTH *"Bots understand language."*

TRUTH They **map** language. A script bot matches patterns to canned replies; a smart bot predicts likely word sequences from statistics. Neither has a referent — no world the words point at. The proof is in T10-08: swap two synonyms a child handles effortlessly and watch the counts barely move. Understanding binds words to the world; models bind words to other words.

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can explain what a machine receives — pixels and channel numbers — and why that makes some illusions work ○○○
- [ ] I can run a 3×3 kernel by hand and say what a feature map records ○○○
- [ ] I can normalise text and defend what normalisation throws away ○○○
- [ ] I can build a bag-of-words table and use TF-IDF logic to find the informative word ○○○
- [ ] I can decide where a script bot suffices and where a smart bot earns its risks ○○○

THINK & INK

A bot I use — school, shop or app. Its honest classification is script/smart because of the moment when …

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter “seeing” and “reading” felt like one magic ability. Now I can name the two number-pipelines — …

CHAPTER 2 CLUES — POCKET THESE

- **A machine sees only numbers** — 0–255 per channel. Fool the numbers and you have fooled the machine.
- **Convolution = slide, multiply-and-add, record.** One kernel, one pattern; a CNN is stacks of them.
- **Normalisation first, counts second, TF-IDF third** — informative means frequent-here-rare-elsewhere.
- **Script bots recite trees; smart bots predict continuations** — neither understands, and the exam question is which the task deserves.

The clue I would tell my family tonight:

Chapter 2 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. Your kernel response for one patch came out NEGATIVE. What did that patch look like — and what does the sign tell the next layer?

2. After normalisation, “urgent” means the same in M1 and M2 — and that is a loss. Name what was lost and one feature that recovers it.

3. The office bot question “will the bus wait if the flight is late?” breaks a script bot at the tree, and a smart bot at the truth. Explain both failures.

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT HOW MACHINES SEE AND READ

HOME LINK *take this page home — no screen needed*

With an adult, find one automated reply in your family’s life — a helpline menu, a delivery bot, a complaint auto-response. Run the test together: ask it one question one step outside its script. Write down the exact failure — and whether it failed like a script bot (lost) or a smart bot (fluent nonsense). Bring the receipt of the conversation to class.

3. Judging a Model

GUIDING QUESTION

Is it any good — good for whom, at what, judged how?

Accuracy is the most quoted number in AI and the least examined. This chapter replaces the single number with a discipline: split the data so the test is honest, count the four fates of every prediction in a confusion matrix, compute precision and recall, and then make the decision that separates professionals from tourists — choosing which metric the **stakes** demand. You will dismantle a "99% accurate" claim with the base-rate trap, and learn the threshold slider that lets a team tune which error it prefers. By the end, you will never again accept a score without asking who was in the test set, who paid for each error, and what the number is hiding.

MISSIONS T10-10 – T10-14
OUTCOMES 10.E1 · 10.S1 **STRANDS** L

Links: Maths — percentages, ratios, conditional probability · Science — screening tests · Civics — accountability

In this chapter

3.1 The honest exam: train/test split and leakage 32
3.2 The confusion matrix: four fates for every prediction 33
3.3 Choosing the metric: precision, recall and the stakes 34
3.4 The base-rate trap: why "99% accurate" can be almost useless 35
3.5 The threshold slider: tuning which error you prefer 37

TERMS YOU'LL DESIGN WITH

train/test split · **confusion matrix** · **precision** · **recall** · **base rate**

WARM-UP · BEFORE YOU READ — ALONE · 2 MIN

A message says: "Our AI detects a rare disease with 99% accuracy." Write the ONE question that, if answered, could make that number nearly worthless. Keep it — Section 3.4 answers it.

DESIGNER'S TOOLKIT — YOU WILL NEED: pencil · ruler · one deck of 20 slips (or chits) per pair · calculator optional

3.1 The honest exam: train/test split and leakage

The first rule of judging: **the model must be tested on examples it has never seen.** Split your data — commonly four-fifths to train, one-fifth to test — and freeze the test set like an exam paper in a sealed envelope. The model may look at the training set as often as it likes; the test set gets one visit, at the end. The sin that ruins this is **leakage**: any way the test information sneaks into training — copying test rows into training, choosing features that secretly contain the answer, or tuning the model after peeking at the test score. A leaked test is not a hard exam; it is a photocopy of the answer key, and every percentage it produces is decoration.

T10-10 Split the Deck

HANDS-ON LAB · PAIR · 15 MIN

Take 20 slips: 12 labelled SICK, 8 HEALTHY. Shuffle. Split 16 / 4 — train / test — **before touching anything else**. Your "model" is one rule you will pick by looking at training slips only.

STEP BY STEP

1. Count your test deck: how many SICK / HEALTHY? Write it here — this is the answer key you must NOT consult:

2. From the 16 training slips, find a rule: "predict SICK if ___". (There is no feature — decide what base rate alone supports: always-SICK? weighted guess?)

3. Apply your rule to the 4 test slips. Predictions vs truth:

THEN ARGUE

a) A classmate peeks at the test slips "just to check the rule is reasonable", then keeps it. Which number is now fake, and why exactly?

b) Another pair accidentally trained on all 20 slips and scored 95%. On the sealed 4, they might score what?

c) Real leakage in the wild: choosing "days in hospital" as a feature to predict "is sick". Why does that feature contain the answer?

The sentence to keep: a test set is a promise you make before you look — and the whole industry of inflated scores is the story of promises made after.

Definition: train/test split

Holding back part of the data so the model is judged only on what it never saw.

Read it three times today — it sticks!

3.2 The confusion matrix: four fates for every prediction

Accuracy counts one thing: right or wrong. But a screening tool that misses disease and a spam filter that blocks your admission letter are both "wrong" — and not remotely the same. The confusion matrix splits the wreckage honestly. For a YES/NO model, every case lands in one of four cells: **true positive** (said yes, was yes), **true negative** (said no, was no), **false positive** (cried yes, was no), **false negative** (said no, was yes). From those four counts, every honest metric is a fraction — and the **choice** among them is where ethics enters the mathematics.

The confusion matrix — where accuracy goes to be judged

A disease-screening model tested on 100 cases. Four fates for every case; five honest numbers out of the same table.

CONFUSION MATRIX 100 test cases

	WHAT THE MODEL SAID: Sick	WHAT THE MODEL SAID: Healthy
TRUTH: Sick	**44** TRUE POSITIVE caught — treated in time	**10** FALSE NEGATIVE missed — sent home sick
TRUTH: Healthy	**6** FALSE POSITIVE false alarm — extra test	**40** TRUE NEGATIVE correctly cleared

Accuracy = (TP + TN)/all = (44+40)/100 = **84%** — one number, hides who got hurt.

Precision = TP/(TP + FP) = 44/50 = **88%** — "when it cries *sick*, how often is it right?"

Recall = TP/(TP + FN) = 44/54 = **81%** — "of all truly sick people, how many did it *catch*?"

F1 = the balance of the two — one score when both errors matter.

CHOOSING IS ETHICS, NOT MATHS A missed disease (FN) costs a life; a false alarm (FP) costs one extra test. Screening should chase **recall**. A spam filter fears blocked real mail — it should chase **precision**. Same table, different duty.

FIG One table, four fates, five honest numbers. The red strip is the decision most courses skip — and this book refuses to.

T10-11 Confusion Matrix from 100 Cases

■ HANDS-ON LAB PAIR · 20 MIN

A uniform-check vision model ran on 100 students. Results: it flagged 50; 44 were genuinely out of uniform, 6 were dressed fine and wrongly flagged. Of the 50 it cleared, 40 were genuinely fine and 10 were out of uniform but missed.

a) Fill the matrix (TP / FP / FN / TN), labelling rows truth, columns prediction — and say what "positive" means in this setup.

b) Compute: accuracy, precision, recall. Show every fraction.

c) The sports teacher proposes the opposite: “flag anyone borderline”. Predict what happens to FP, FN, precision and recall — direction only, no numbers.

d) Who is harmed by each cell of YOUR matrix — name the person in each, and what they lose.

WORD POWER · 19

Definition: **confusion matrix**

The 2×2 table of what a model said versus what was true.

Read it three times today — it sticks!

3.3 Choosing the metric: precision, recall and the stakes

Precision measures the yes-column: of everything the model flagged, how much was real. **Recall** measures the yes-truth: of everything real, how much the model caught. They pull against each other — catch harder and you gain recall while false alarms erode precision; relax and precision climbs while misses grow. There is no universal winner; there is only **the cost of each error in this context**. Screening for a lethal disease: a missed case can kill, a false alarm costs one more test — chase recall. Blocking messages: a blocked admission letter is a disaster, a missed spam is a shrug — chase precision. A cheating detector chasing recall accuses innocents; a fraud detector chasing precision misses frauds. The metric is the ethical position, written as arithmetic.

T10-12 Precision vs Recall: Which Matters?

GROUP · 20 MIN

Three deployments, three debates. For each: **which metric should the team chase — and what error, in plain words, are they choosing to tolerate?**

Deployment	Metric to chase	The error we accept — and why
1 · Cancer screening tool for a village health camp		
2 · Spam filter for the principal's office inbox		
3 · Exam cheating detector flagging papers for review		

The debate your teacher will love: one member argues the OPPOSITE choice for row 1 — under what real-world constraint (medicine shortage? stigma? cost of the confirmatory test?) could their argument win?

The sentence to keep: a metric is not a preference — it is a written record of whom the design is willing to hurt, and how often.

WORD POWER · 20

Definition: **precision**

Of everything flagged, the fraction that was real — the cost of false alarms.

Read it three times today — it sticks!

WORD POWER · 21

Definition: **recall**

Of everything real, the fraction that was caught — the cost of misses.

Read it three times today — it sticks!

3.4 The base-rate trap: why "99% accurate" can be almost useless

Here is the trap that catches newsreaders, investors and principals alike. A rare disease affects 1 person in 1,000. A test is "99% accurate" — it is right about everyone, sick or healthy, 99 times in 100. Test 1,000 people: about 10 are sick. The test catches nearly all of them — but its 1% error

rate also accuses about 10 **healthy** people. Look at the positives: roughly 10 true, roughly 10 false. Half the alarms are wrong. The base rate — how rare the condition is to begin with — quietly governs what a positive **means**. The general law: with rare conditions, even excellent tests produce mostly false alarms, and the rarer the condition, the worse the odds behind the headline.

T10-13 "99% Accurate"? The Base-Rate Trap

■ QUICK WIN ALONE · 15 MIN

Work the numbers. Disease affects 1 in 1,000. Test: 99% accurate both ways (it correctly reports 99% of the sick as sick, and 99% of the healthy as healthy).

a) In a tested population of 10,000: how many are truly sick? Truly healthy?

b) How many true positives? How many false positives? (Show the two multiplications.)

c) Of all people who test positive, what fraction is actually sick? Write it as a fraction and a percentage.

d) Rewrite the advert in one honest sentence: "99% accurate — and yet …"

e) Now the disease doubles to 2 in 1,000. What happens to your answer to (c) — and what does that tell you about base rates?

Definition: base rate

How common a condition is before any test — the prior that quietly governs every positive.

Read it three times today — it sticks!

MYTH BUSTER

MYTH *"Accuracy is enough."*

TRUTH Accuracy blends four different fates into one number — and hides who paid. A model that says "healthy" to everyone scores 99.9% on a population where the disease affects 1 in 1,000, while missing every single case. After this chapter you read four numbers, not one: the confusion matrix, then the metric the stakes choose.

3.5 The threshold slider: tuning which error you prefer

Most models do not answer "yes" or "no"; they output a **score** — 0.82, 0.47, 0.63 — and a threshold line converts scores into decisions: above the line, flag; below, pass. Slide the threshold down and you flag more: recall rises, precision falls. Slide it up: precision rises, recall falls. The slider is where the Chapter 3 argument becomes an engineering act — the screening tool and the spam filter could share the same model and differ only in where the line sits. Choosing a threshold **is** choosing your false-positive budget, in public, with numbers attached.

T10-14 Threshold Slider on Paper

HANDS-ON LAB PAIR · 15 MIN

Ten patients scored by one model: A 0.94, B 0.88, C 0.71, D 0.66, E 0.55, F 0.49, G 0.38, H 0.30, I 0.22, J 0.11. Truth: A, B, C, D, G are sick; the rest healthy.

Threshold	Flagged as sick	TP · FP · FN · TN	Precision	Recall
0.9				
0.6				
0.3				

a) Complete the table. One row as a worked example for your partner, the rest verified by a second pencil.

b) You are the health-camp lead from T10-12 row 1. Mark your chosen threshold with a star — and defend it against the pair who chose differently.

c) The principal (T10-12 row 3's world) borrows your slider. Which end do they move it — and what do they write in their justification?

WORD POWER · 23

Definition: threshold

The score line that turns a model's number into a yes-or-no decision.

Read it three times today — it sticks!

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can split data so the test stays honest — and name the leakages that fake a score ○ ○ ○
- [ ] I can build a confusion matrix from raw results and compute accuracy, precision and recall from it ○ ○ ○
- [ ] I can choose a metric for a context and defend it with the cost of each error ○ ○ ○
- [ ] I can dismantle a "99% accurate" headline using the base-rate trap ○ ○ ○
- [ ] I can predict what moving a threshold does to precision and recall — before computing it ○ ○ ○

THINK & INK

A headline score I have believed this year. After this chapter, the question I now bring to any such number is …

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter, a good percentage felt like a verdict. Now I treat every score as the beginning of …

CHAPTER 3 CLUES — POCKET THESE

- **Freeze the test set first** — a promise made before looking. Leakage is any backdoor from test to training.
- **Four fates, not one:** TP, FP, FN, TN. Every honest metric is a fraction of this matrix.
- **Precision buys trust in alarms; recall buys coverage of the truth** — the stakes pick which one you chase.
- **Rare condition + good test = mostly false alarms.** The base rate quietly governs what a positive means.

The clue I would tell my family tonight:

Chapter 3 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. The uniform model's recall rose from 88% to 96% overnight. Name two things that could explain it — one honest, one that should get the model suspended.

2. A village camp screens 5,000 people for a condition affecting 1 in 200, with a 98%-accurate test. Roughly how many false alarms — and why does the camp need to plan for them?

3. Why is "we chose recall" NOT the same as "we chose the best metric"? Answer with the word 'cost' and one real stakeholder.

THE METRIC CHEAT CARD — CUT IT OUT MENTALLY, KEEP IT FOREVER

When the harm is…	…the metric to chase	The sentence I will say in the meeting
a miss ruins a life (disease, smoke alarm, fraud loss)	**recall** — catch nearly everything	"We accept more false alarms; here is the budget for them."
a false alarm ruins a life (accusation, blocked letter, locked account)	**precision** — almost never cry wolf	"We accept some misses; here is the human backstop for them."
both errors cost real money, no lives	**F1** — the balanced default	"We optimise the balance and re-check each quarter."

And under every card, the Chapter 3 promise: the test set was sealed FIRST, the base rate was checked, and somebody owns each of the four cells by name.

RE-RATE YOURSELF — CHAPTER 3 EDITION

Split the deck, sealed first: E · D · P · A
Choose the metric for the stakes: E · D · P · A
Build the matrix, four fates: E · D · P · A
Dismantle a "99%" headline: E · D · P · A

Any row you would not circle P or A yet — name the mission you will re-run before the capstone:

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT JUDGING A MODEL

HOME LINK *take this page home — no screen needed*

With an adult, find one health, weather or safety claim in this week's news that quotes an accuracy or a percentage. Together, apply the base-rate question: rare or common, tested on whom, and what does a positive actually mean here? Write the one-line verdict you would print under the headline — and bring it for the Chapter 4 wall.

4. Ethics, Governance & Society

GUIDING QUESTION

Who decides — and who answers when the decision harms?

Chapters 1–3 gave you the machinery; this chapter gives it a conscience and a courtroom. You will run a medical-triage tool through the four bioethics principles that doctors have argued with for decades, conduct a full algorithm audit — bias, transparency, accountability, each with evidence — map your privacy rights against your responsibilities, and face generative AI as the integrity question it truly is: what may a machine write in your name, and what must never leave your hands? The chapter's quiet thesis: "legal" and "ethical" are different words, and the space between them is where every professional you will ever meet does their real work.

MISSIONS T10-15 – T10-18
OUTCOMES 10.R1 · 10.R2 **STRANDS** R W

Links: Biology — medical ethics · Civics — rights & governance · Economics — labour markets · English — reasoned argument

In this chapter

4.1 Four principles, one case: the ethics frame with teeth 42
4.2 The algorithm audit: bias, transparency, accountability 43
4.3 Privacy: your rights and your responsibilities 44
4.4 Generative AI and integrity — plus the career map 45

TERMS YOU'LL DESIGN WITH

accountability · transparency · data minimisation · integrity

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

A hospital deploys a triage tool that cuts waiting times by 30% — and quietly ranks one neighbourhood's patients lower. Which single word names what must happen next? Keep your answer for Section 4.2.

DESIGNER'S TOOLKIT — YOU WILL NEED: pencil · sticky notes · your Chapter 3 metric choices · one news item about an AI decision

4.1 Four principles, one case: the ethics frame with teeth

Medicine arrived at its four principles after decades of real harm: **autonomy** (people choose for themselves), **beneficence** (actively do good), **non-maleficence** (do no harm), **justice** (fair shares of benefit and burden). AI inherits all four the day a model starts ranking, triaging or recommending for people. The frame's power is its honesty: a design can satisfy two principles and fail the others, and the failure is specific enough to fix. Run the four questions **before** deployment — after deployment, you are not designing any more; you are apologising.

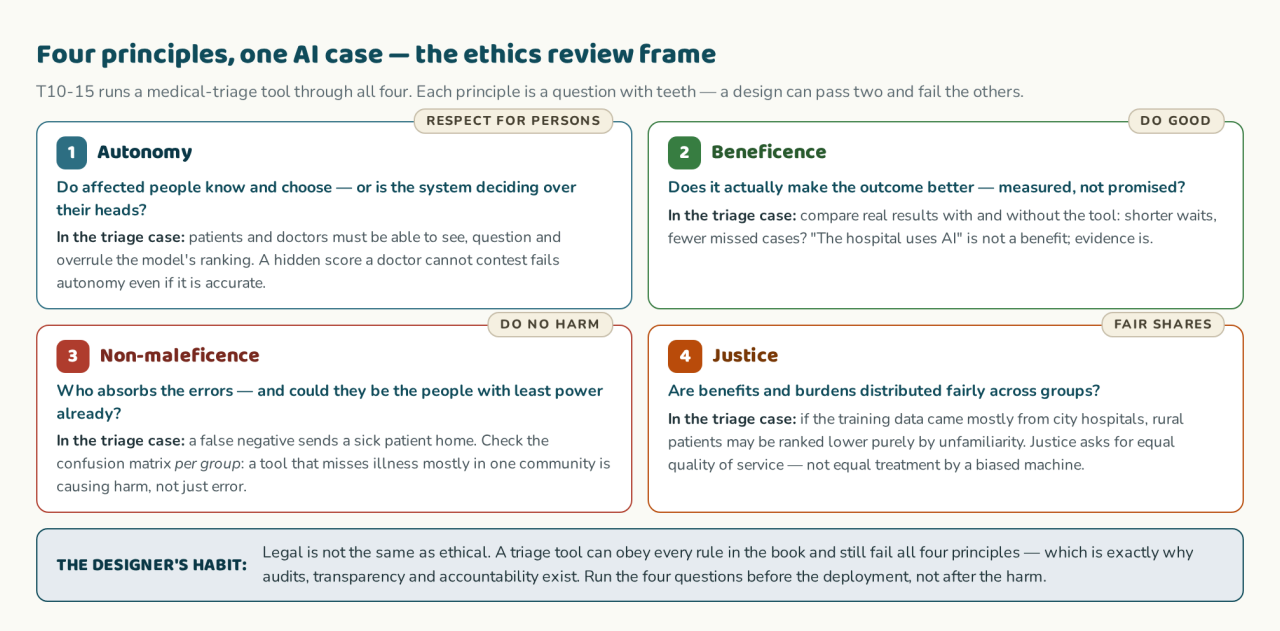


FIG The four principles as an instrument panel. Each needle must move; two green dials do not make a safe flight.

T10-15 Four Principles Case

GROUP · 25 MIN

The case: a city hospital trials a triage tool that reads patient files and ranks who should be seen first. Waiting time drops 30% overall. Complaints say patients from the old town clinic rank lower, the tool's reasoning is not shown to doctors, and nobody at the hospital can say who owns its errors.

Principle	Verdict: pass / strained / fails	The specific evidence from the case
Autonomy		
Beneficence		
Non-maleficence		
Justice		

a) Your group's single strongest fix, chosen from ALL four rows — and which principle it repairs.

b) The hospital says "overall benefit is 30%, so deploy". Which principle is that argument using as a shield — and what does the frame answer?

c) Name the one group whose voice is missing from this case entirely — and the question you would put to them.

WORD POWER · 24

Definition: **accountability**

A named person or body that answers for a system's decisions — in writing, in advance.

Read it three times today — it sticks!

4.2 The algorithm audit: bias, transparency, accountability

An audit is what professionals do instead of arguing. **Bias**: compare outcomes across groups — does the model serve one community's accents, faces, postcodes measurably worse? **Transparency**: can the people affected find out what the system considers and why — at least the factors, if not the code? **Accountability**: is there a named owner, an appeal route, a log? A tool can be accurate and still fail an audit — that is the gap between the syllabus question and the news story. Auditing is a skill with a shape: claim, evidence, verdict, fix — the same shape your lab reports have used for years.

T10-16 Algorithm Audit Report

HANDS-ON LAB GROUP · 30 MIN

Audit ONE system your group knows (a recommendation feed, a helpline bot, a school CCTV alert, an exam proctoring tool — invented details allowed, real names not required).

BIAS — WHO SERVES WORSE? *Name the groups; state the measurable difference you suspect; say how you would test it.*

TRANSPARENCY — CAN THEY SEE IN? *What can an affected person find out about the decision — factors, data, appeal? What is hidden?*

ACCOUNTABILITY — WHO ANSWERS? *Named owner? Appeal route? Log of decisions? Write who SHOULD own it, by role.*

VERDICT & ONE FIX *Ship / fix-first / stop. The single change that most improves the audit — and who must approve it.*

Present in 90 seconds: your verdict, your fix, and the one question you could not answer from outside the system. Unanswerable questions go on the class wall — they are Chapter 5 material.

WORD POWER · 25

Definition: **transparency**

Whether affected people can see what a system considers and why — at least the factors.

Read it three times today — it sticks!

4.3 Privacy: your rights and your responsibilities

Privacy is not secrecy; it is **control over your own story**. A working set of rights, in plain language: to know what is collected; to reach it and correct it; to know whom it was shared with; to withdraw; and to have it deleted when the purpose ends. India's data-protection law (the DPDP Act, 2023) gives these ideas legal form — verify the current rules with your teacher before quoting section numbers in an exam, because this print may outlive a regulation. The

counterweight is **data minimisation**, the designer's discipline: collect only what the purpose needs, keep it only as long as the purpose lives. Every field you delete is a field that can never leak, never be subpoenaed, never be reused against its subject.

T10-17 Privacy Rights Scenarios

PAIR · 15 MIN

For each scenario: which right is in play — and what should happen next?

Scenario	Right in play	What should happen
A learning app keeps student error-logs 'forever for research', with no delete option		
A class group photo is fed to a face-tagging service without asking the students		
A delivery app shares your exact address with three 'partners' you never named		
You ask a platform what data it holds; it replies with an advert		

The designer's turn: your Chapter 5 brief will collect data from real people. Write your **minimisation list** now — every field you will NOT collect, and the one sentence of consent you will use.

WORD POWER · 26

Definition: **data minimisation**

Collect only what the purpose needs; keep it only while the purpose lives.

Read it three times today — it sticks!

4.4 Generative AI and integrity — plus the career map

Generative AI writes essays, solves sums, drafts applications — and this year it sits in every exam hall in the country. Integrity is the design question, not the policing one: **what is**

schoolwork for? If the purpose is the judgement you build while doing it, then outsourcing the judgement empties the task while leaving it technically complete. The honest line most professionals draw: AI may draft, suggest and explain; the student must be able to defend every claim it produced, and must say what they used. That same honesty is the career skill. Roles are reshaping rather than vanishing — data work, model evaluation, AI oversight, domain-expert-plus-tools. Chapter 5 closes the series with your personal map.

HUMAN VS MACHINE **Judgement vs optimisation** HUMAN AI

A model minimises the loss it is given; a designer decides what is worth minimising — and what it may cost. That decision is the human's, always.

THE HUMAN WAY

You can change the goal mid-flight: notice that recall matters more than accuracy here, weigh who bears each error, and refuse a metric that hides harm.

THE MACHINE WAY

The optimiser reaches whatever target it is given — including a harmful one — with perfect discipline and zero awareness that a choice was ever made.

Where you would overrule a perfect metric in your capstone — and the principle you would cite:

A screening tool hits 99% accuracy and the room cheers. Write the designer's next question:

Name one decision in this book's own design — missions, order, honesty — that an optimiser would never have made:

T10-18 Future Jobs & Skills

ALONE · 15 MIN

a) Pick three roles you might want at 25 (engineer, doctor, teacher, designer, civil servant, farmer-entrepreneur, anything). For each, write: which parts of the job a model could draft — and which parts stay human because they carry judgement or responsibility.

b) Name the one skill that grows MORE valuable in all three of your rows. Defend the claim in one sentence.

c) The honest risk: name one task in your dream role that could shrink — and what you would do about it **on purpose**.

d) Your integrity line for this exam year: complete "I will use AI to help me …, and I will never let it …"

MYTH BUSTER

MYTH *"If it is legal, it is ethical."*

TRUTH Law is the floor, not the ceiling — it codifies what a society has agreed to punish, slowly and after the harm. The four principles ask what you should do while the law catches up: the triage tool in T10-15 could pass every current regulation and still fail justice. Every audit you run in life will find its hardest questions in the legal-but-wrong zone.

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can run the four principles over an AI case and say exactly where each one bites ○○○
- [ ] I can audit a system for bias, transparency and accountability — with evidence, not vibes ○○○
- [ ] I can list my privacy rights and apply data minimisation to a design of my own ○○○
- [ ] I can state an integrity line for generative AI that I could defend to my examiners ○○○
- [ ] I can point at the legal-ethical gap in a real news case and name what should change ○○○

THINK & INK

A system I use that is probably legal and probably unfair. If I audited it (T10-16), the first thing I would find is ...

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter "ethics" felt like opinions. Now it has four questions, an audit shape and a verdict — and the part I now see everywhere is ...

CHAPTER 4 CLUES — POCKET THESE

- **Four principles:** autonomy, beneficence, non-maleficence, justice. Run all four; two green dials do not make a safe flight.
- **Audit shape:** claim, evidence, verdict, fix — for bias, transparency and accountability in turn.
- **Privacy is control, and minimisation is the designer's discipline** — every field you delete is a field that can never leak.
- **Integrity line:** AI may draft and explain; you must be able to defend every claim — and say what you used.

The clue I would tell my family tonight:

MY INTEGRITY LINE — FINAL WORDING

You wrote a first draft on the journey-map page. Now — one chapter wiser, and with the four principles in hand — write the version you will actually keep:

Chapter 4 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. The hospital calls its triage tool ‘a doctor-support aid’ to relax the rules. Which principle does that label touch first — and does the label survive contact with the case?

2. Give one privacy right the learning app violated and ONE action a student could reasonably take this week.

3. Why does “the law allows it” end an argument in a courtroom but not in a design review? Answer with the four-principles frame.

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT ETHICS AND GOVERNANCE

HOME LINK *take this page home — no screen needed*

With an adult, read ONE page of a real privacy policy or app permission screen — any service your family uses. Together, find: one field collected that the purpose does not need (minimisation miss), and one right you cannot locate (knowledge gap). Bring the page; Chapter 5's consent clause starts there.

5. Capstone: The Responsible AI Design Brief

GUIDING QUESTION

Can you design AI that survives its own audit?

Four books of tools converge on one document. Your team will design a complete AI system on paper — scoped problem, consented and audited data plan, chosen model family, honest evaluation with a defensible metric, deployment limits, an ethics gate that can stop the project, and a pitch that names its own doubts before the judges do. Then you will defend it to a Board of Judges made of your classmates, and finally turn the map toward yourself: where this way of thinking meets the subjects, courses and careers you are about to choose. This is the assignment the whole series was quietly rehearsing.

MISSIONS T10-19 – T10-21
OUTCOMES 10.D1 · 10.C1 **STRANDS** U D L W R

Links: all subjects — the brief is interdisciplinary by design · English — the pitch is reasoned writing · Economics — budgets and trade-offs

In this chapter

5.1 The brief: seven stages, two gates 51
5.2 The Board of Judges 53
5.3 Where next: your subject and career map 54

TERMS YOU'LL DESIGN WITH

design brief **ethics gate** **deployment limits** **pitch**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

Write the problem your team is tempted to build for — in one messy sentence. You will sharpen it at Stage 1 until it is boring, precise and un-fakeable.

DESIGNER'S TOOLKIT — YOU WILL NEED: your Chapter 1 family map · Chapter 3 metric choices · Chapter 4 minimisation list · A3 paper or two notebook pages

5.1 The brief: seven stages, two gates

The pipeline below is the whole series in one line. Stages 1–4 are the engineering spine; Stage 5 decides who stays in the loop; Stage 6 is the ethics review with the power to say no; Stage 7 is where you defend it. The two gates matter more than the stages: **Gate A** stops bad data before a model exists, and **Gate B** stops unaudited deployments before real people meet them. Professional teams that skip gates do not move faster — they just schedule their failure for after the launch.

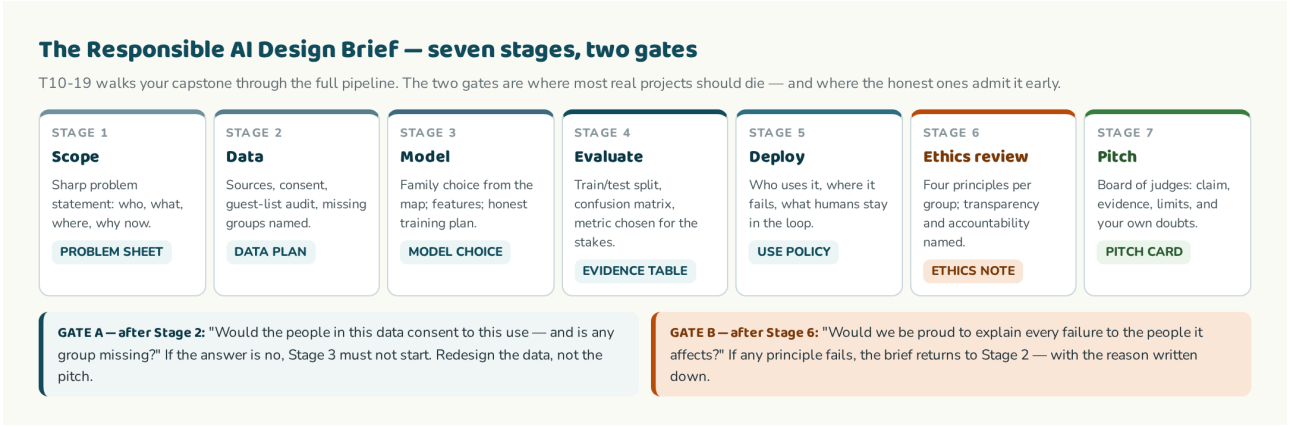


FIG Your capstone in one strip: seven stages, two gates. Stage 6 exists so that "stop" is a legal move in your own process.

HOW REAL TEAMS RUN THIS BRIEF — THE THREE-PERIOD RHYTHM

PERIOD 1 · SCOPE + DATA
Stages 1–2 and Gate A. End only when the consent sentence and the missing-groups list are written — not when time is up.

PERIOD 2 · MODEL + EVALUATION
Stages 3–4. The metric argument out loud, in roles: one argues recall, one precision, one is the stakeholder who pays.

PERIOD 3 · DEPLOY + REVIEW + GATE B
Stages 5–6. Then rehearse the pitch against the five board questions — the team that rehearses its limits owns the room.

T10-19 Responsible AI Design Brief

■ HANDS-ON LAB GROUP · 3–4 periods MIN

One problem your team genuinely cares about — school, neighbourhood, town. Build the brief in two parts.

PART A · STAGES 1–4 + GATE A

1 · Scope — the boring, precise statement: who, what, where, why now.

2 · Data — sources, consent sentence, guest-list audit, named missing groups, minimisation list.

3 · Model — family + branch + job, chosen from your map, with the reason.

4 · Evaluate — split, metric chosen FOR THE STAKES, expected confusion-matrix shape.

GATE A — WOULD THE PEOPLE IN THIS DATA CONSENT? IF NO, STOP AND REWRITE THE DATA PLAN.

PART B · STAGES 5–7 + GATE B

5 · Deploy — who uses it, where it fails first, which humans stay in the loop.

6 · Ethics review — four principles, one line each; your audit verdict: ship / fix-first / stop.

7 · Pitch — the claim, the evidence, the limits, and what would change your mind.

GATE B — WOULD YOU EXPLAIN EVERY FAILURE TO THE PEOPLE IT AFFECTS? IF NOT, BACK TO STAGE 2, IN WRITING.

Budget line (every real brief has one): what does maintaining this cost — data refreshes, audits, appeals? Name the cheapest thing you would cut FIRST when money runs out, and what that cut does to your metric.

5.2 The Board of Judges

The pitch is reasoned writing under pressure. Five minutes per team: the claim, the evidence, the limits — and the board's job is not to be nice; it is to find the assumption the team has not tested. Every question lands on one of the seven stages, which is why you drew them.

T10-20 Board of Judges Pitch

CLASS · 25 MIN

For the pitching team (5 minutes): the problem in one sentence · the metric you chose and the error you accept · the gate you almost failed, and what you changed · the limit you discovered yourself.

For each judge — one question from a different stage:

Judge	Stage attacked	The question (write it before they present)
Judge 1	Stage 1 · Scope	
Judge 2	Stage 2 · Data	
Judge 3	Stage 4 · Metric	
Judge 4	Stage 6 · Ethics	

The verdict rule: the board may send a team back through a gate, but every "back" must name the stage and the reason. Write the board's verdict here — including the assumption the team had NOT tested:

5.3 Where next: your subject and career map

The series ends by turning the map toward you. The AI-literate Class 10 student does not have to become an engineer; they have to become the person in the room who can ask the seven questions and read the four numbers. Every board subject you are choosing feeds that person: mathematics feeds the metrics, biology feeds the screening cases, civics feeds the governance debates, languages feed the text systems, economics feeds the budgets.

T10-21 Where Next? Subject & Career Map

QUICK WIN · ALONE · 15 MIN

a) From your 10 "I can" statements (front of this book), star the TWO you are already **Proficient** at. Evidence, not feeling — name the task that proves each.

b) Circle the ONE you most want at **Advanced** by year-end. What would you design or audit to get there?

c) Your subject choices for the next two years — and the ONE chapter of this series each subject will inherit from you.

d) One line to the designer you are becoming: what will you refuse to build, whatever the salary?

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can run the seven-stage brief from scope to pitch without skipping a gate ○○○
- [ ] I can defend a metric choice against a board that disagrees with me ○○○
- [ ] I can name the assumption my own design rests on — and what evidence would move it ○○○
- [ ] I can map my starred skills to subjects and careers honestly, with evidence ○○○
- [ ] I can state what I will refuse to build, and why that is a design decision, not a slogan ○○○

THINK & INK

Five books, one road: the thing I can DO now that I could not do in Class 6 is …, and the thing I still want to own by Class 12 is …

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before the series I thought AI was products. Now I know it is decisions wearing products — and the decision I feel most ready to make is …

CHAPTER 5 CLUES — POCKET THESE

- **Seven stages, two gates** — the gates are where honest projects stop, fix, and restart in writing.
- **A pitch names its own limits** before the judges find them — credibility is a design material.
- **The verdict must name the stage and the reason** — "back to Stage 2" is a sentence, not a feeling.
- **Your map forward** is starred evidence, not hopes — two Proficient claims with proof, one Advanced target with a plan.

The clue I would tell my family tonight:

Chapter 5 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. A rival team's brief has brilliant metrics and no consent sentence. Which gate stops it — and in one line, what the team must produce to pass?

2. The board cannot agree on your verdict. Write the ONE piece of evidence that would settle it — and which stage it belongs to.

3. Why is "what would change my mind" a Stage 7 requirement and not a Stage 4 one? Answer with the words 'claim' and 'stakeholder'.

WHAT EVERY BOARD ASKS — THE FIVE QUESTIONS OF EVERY REVIEW ROOM YOU WILL EVER ENTER

- **"What exactly is the problem — and who chose it?"** (Stage 1)
- **"Where did this data come from, and who consented?"** (Stage 2)
- **"Which error does your metric accept, and who pays it?"** (Stage 4)
- **"Show me the failure you found yourself — not the one we found."** (any stage)
- **"What would make you stop your own project?"** (Gate B)

Tick the ones your pitch survived today. The unticked ones are not failures — they are your revision list, and every professional review room in the country runs on exactly these five.

MY CASE JOURNAL — THE LAST ENTRY OF THE SERIES: WHAT I WILL DESIGN, AND WHAT I WILL REFUSE

My AI words

The twenty-four words on the final case — each one met through an experience, never before.

1 rule-based system
A system whose every behaviour follows rules written entirely by people.
USE IT IN MY OWN SENTENCE:

2 supervised learning
Learning from examples that come with correct answers attached.
USE IT IN MY OWN SENTENCE:

3 unsupervised learning
Finding structure in examples that carry no labels at all.
USE IT IN MY OWN SENTENCE:

4 reinforcement learning
Learning a strategy by acting, then adjusting to rewards and penalties.
USE IT IN MY OWN SENTENCE:

5 classification
A learning job whose output is a category from a fixed list.
USE IT IN MY OWN SENTENCE:

6 regression
A learning job whose output is a number on a scale.
USE IT IN MY OWN SENTENCE:

7 clustering
A learning job that finds groups nobody named in advance.
USE IT IN MY OWN SENTENCE:

8 association
A learning job that finds items that keep appearing together.
USE IT IN MY OWN SENTENCE:

9 weight
The adjustable number a neuron multiplies an input by — a connection's strength.
USE IT IN MY OWN SENTENCE:

10 layer
A row of neurons whose outputs feed the next row; depth comes from stacking.
USE IT IN MY OWN SENTENCE:

11 pixel
One position in an image grid, holding a brightness number — three for colour.
USE IT IN MY OWN SENTENCE:

12 grayscale
An image where each pixel holds a single brightness number from 0 to 255.
USE IT IN MY OWN SENTENCE:

13 kernel
The small grid of weights slid across an image — one learned pattern-detector.
USE IT IN MY OWN SENTENCE:

14 feature map
The grid of responses one kernel produces — where its pattern fired.
USE IT IN MY OWN SENTENCE:

15 normalisation
Cleaning text so machines can count it: lowercase, no punctuation, stems chopped.
USE IT IN MY OWN SENTENCE:

16 bag-of-words
A table of word counts per message — order lost, frequency kept.
USE IT IN MY OWN SENTENCE:

17 TF-IDF
A score that lifts words frequent here but rare elsewhere — the informative ones.
USE IT IN MY OWN SENTENCE:

18 train/test split
Holding back part of the data so the model is judged only on what it never saw.
USE IT IN MY OWN SENTENCE:

19 confusion matrix
The 2×2 table of what a model said versus what was true.
USE IT IN MY OWN SENTENCE:

20 precision
Of everything flagged, the fraction that was real — the cost of false alarms.
USE IT IN MY OWN SENTENCE:

21 recall
Of everything real, the fraction that was caught — the cost of misses.
USE IT IN MY OWN SENTENCE:

22 base rate
How common a condition is before any test — the prior that governs every positive.
USE IT IN MY OWN SENTENCE:

23 threshold
The score line that turns a model's number into a yes-or-no decision.
USE IT IN MY OWN SENTENCE:

24 accountability
A named person or body that answers for a system's decisions — in writing, in advance.
USE IT IN MY OWN SENTENCE:

THE DESIGNER'S OATH

Machines **classify, estimate, cluster, flag and score** — people **scope the problem, choose the data, pick the metric, own the errors and sign the gates**. When a system speaks, I will ask my questions: **How do I know? What is missing? Who made this? Who is missing? What is the trick here? Who is accountable? What would change my mind?**

ANSWER HINTS — FOR CHECKING, NEVER FOR COPYING

T10-01: the lift is rule-based to its bones; the chess self-improver and the essay chatbot are learning-based — but only the chess program is reinforcement (it plays the world for points); the kirana rule and the lift controller are the pair most worth arguing. · **T10-03:** tuition marks = supervised; bus rider groups = unsupervised; drone = reinforcement; X-rays = supervised; singer pairs = unsupervised; vacuum = reinforcement. · **T10-04:** rice+dal appears in bills 1,2,3,7 — confidence 4/6 ≈ 67% of rice bills; the app is wrong in 2 of 6 rice cases ≈ 33% of the time; bill 4 (bread, jam) shows how fast small samples break.

MORE HINTS — FOR THE STICKY MISSIONS

T10-05: your two hand-computed patches must match the worked patch's arithmetic — if the class disagrees on one cell, redo the nine multiplications aloud in turn; the kernel never moves half a cell. · **T10-08:** "free" is likely your highest document-frequency word AND a weak discriminator if it appears everywhere — TF-IDF intuition: the informative words are the ones one message cannot stop using. · **T10-10:** with 12 sick in 20 slips, "always SICK" on a random 4-test scores about 60% — and a peeked test scores exactly 100% and exactly nothing. · **T10-11:** TP=44, FP=6, FN=10, TN=40; accuracy 84%, precision 44/50 = 88%, recall 44/54 ≈ 81%. · **T10-13:** of 10,000: 10 sick, 9,990 healthy; TP ≈ 10, FP ≈ 100; a positive is truly sick only 10/110 ≈ 9% — the advert survives, the meaning does not.

MY HINT LOG — WHICH MISSIONS NEEDED A PEEK?

Hints are training data for you. Tick the mission code each time a hint unstuck you — a mission that needed three hints deserves one more retry next week, not three more hints:

- ☐ T10-__ needed a hint
- ☐ T10-__ needed a hint
- ☐ T10-__ needed a hint
- ☐ T10-__ needed a hint

The series complete

Five cases, five levels, one habit of mind. This page is the whole journey in your own handwriting.

My series so far — five cases, one designer

CLASS 6 · NOTICE
I learned to **spot** learning machines and ask: how do I know?

CLASS 7 · SORT & PREDICT
I learned to **train and test** machines — and how examples fool them.

CLASS 8 · BUILD THE CYCLE
I learned to **plan, audit and defend** an AI before it is built.

CLASS 9 · REASON
I learned to **compute the mechanisms** — and verify every claim.

CLASS 10 · EVALUATE & DESIGN
I learned to **judge with evidence** and **design with a conscience**.

The habit I am proudest of now:

Where I used my questions — the tally of five finished cases

My question	Where I asked it for real (at home, in class, online)
How do I know?	
What is missing? / Compared to WHAT?	
Who made this? / Who is missing? / Who is accountable?	
What is the trick here? / What would change my mind?	

Seven questions is the whole method. The two newest ones are the ones adults get paid to ask.

THE ROAD AFTER THIS SERIES

There is no Class 11 handout — by design. What comes next is **using** the method where you sit: board subjects, first projects, first jobs. The seven questions travel; the confusion matrix travels; the gates travel. Wherever you meet a system that ranks people, you are now the one in the room who can ask what it hides and who answers for it. Keep every notebook; they are your evidence — of the systems you will one day be trusted to build.

How do I know? · What is missing? · Who made this? · Who is missing? · What is the trick here? · Who is accountable? · What would change my mind?

THIS CASE FILE BELONGS TO THE EVIDENCE

Designer: _____ Class & division: _____ School year: _____

Keep this book — of all five, this is the one to show at your first interview.

DESIGNER'S NOTES — anything else this final case taught me that did not fit anywhere

THE CLUE I WOULD PUT IN A BOTTLE — for whoever finds this series in ten years

If a machine from 2036 reads these pages, here is what I want it to know about the students who learned to judge it on paper:

My progress

Shade honestly — designers never fake evidence.

MY OUTCOME TRACKER — RATE YOURSELF: E EMERGING · D DEVELOPING · P PROFICIENT · A ADVANCED

- ☐ 10.U1 — place products on the model-family map: rule-based, supervised, unsupervised, reinforcement
- ☐ 10.L1 — run a neural network by hand and explain what weights and layers do
- ☐ 10.E1 — split data honestly; build a confusion matrix; compute precision, recall; choose the metric for the stakes
- ☐ 10.V1 — apply a 3×3 kernel by hand and sketch what convolution layers do
- ☐ 10.N1 — normalise text, build a bag-of-words, and tell script bots from smart bots
- ☐ 10.S1 — explore a small dataset's summary values, patterns and limits, and state all three
- ☐ 10.R1 — run the four principles and an audit over a real AI case, with a verdict and a fix
- ☐ 10.R2 — state my privacy rights and design with minimisation; use generative AI with integrity
- ☐ 10.D1 — design an AI project end-to-end on paper, gates included, and pitch it
- ☐ 10.C1 — map my own interests to AI-related study paths and careers, with evidence

MY CASE FILES — SHADE A STAR PER CHAPTER FINISHED *1 Models Explained · 2 See & Read · 3 Judging a Model · 4 Ethics & Society · 5 Design Brief*



MY OUTCOME REFLECTION — read your own ratings before you sign

The rating I can defend with evidence: _____ — the one that still needs work: _____

My plan for the one that needs work (which mission I will re-run, and when):

PRATIMAI · AI HANDOUTS · CLASS 10

My AI Designer's Certificate

This certifies that designer

_____

DESIGNER NAME

has completed the five-case series — from **"AI Around Me"** to **"Decide, Evaluate, Design"** — judged models with evidence, audited systems with principles, designed with both gates shut, and promised to keep asking: **How do I know? · What is missing? · Who made this? · Who is missing? · What is the trick here? · Who is accountable? · What would change my mind?**

DATE	DESIGNER'S SIGNATURE	GUIDE'S SIGNATURE
_____	_____	_____

The series is complete: five levels, five badges. Wherever you meet a system that ranks people, the questions travel with you.

My badge shelf — five years, five levels

1	2	3	4	5
NOTICE	**SORT & PREDICT**	**BUILD THE CYCLE**	**REASON**	**EVALUATE & DESIGN**
CLASS 6	CLASS 7	CLASS 8	CLASS 9	CLASS 10
badge earned ✓	*badge earned ✓*	*badge earned ✓*	*badge earned ✓*	*badge earned ✓*

One line per badge — what each year changed in how I think: